Pathogenesis and management of Brugada syndrome

Juan Sieira[1,2], Gregory Dendramis[1,3] and Pedro Brugada[1]

Abstract | Brugada syndrome is an inherited disease characterized by an increased risk of sudden cardiac death owing to ventricular arrhythmias in the absence of structural heart disease. Since the first description of the syndrome >20 years ago, considerable advances have been made in our understanding of the underlying mechanisms involved and the strategies to stratify at-risk patients. The development of repolarization–depolarization abnormalities in patients with Brugada syndrome can involve genetic alterations, abnormal neural crest cell migration, improper gap junctional communication, or connexome abnormalities. A common phenotype observed on the electrocardiogram of patients with Brugada syndrome might be the result of different pathophysiological mechanisms. Furthermore, risk stratification of this patient cohort is critical, and although some risk factors for Brugada syndrome have been frequently reported, several others remain unconfirmed. Current clinical guidelines offer recommendations for patients at high risk of developing sudden cardiac death, but the management of those at low risk has not yet been defined. In this Review, we discuss the proposed mechanisms that underlie the development of Brugada syndrome and the current risk stratification and therapeutic options available for these patients.

[1]*Heart Rhythm Management Centre, UZ Brussel-VUB, Brussels, Laarbeeklaan 101, 1090 Brussels, Belgium.*
[2]*Cardiology Department, University Hospital Erasme, Route de Lennik 808, 1070 Brussels, Belgium.*
[3]*Cardiovascular Division, University Hospital "Paolo Giaccone", Via Del Vespro 127. 90127 Palermo, Italy.*

Correspondence to P.B.
pedro@brugada.org

doi:10.1038/nrcardio.2016.143
Published online 15 Sep 2016

First described in 1992, Brugada syndrome is an inherited disease characterized by an increased risk of sudden cardiac death (SCD) in the absence of structural heart anomalies. Brugada syndrome has been reported to be responsible for 5–40% of sudden deaths in patients without structural heart disease and is an important cause of death in individuals aged <40 years[1–6]. The global prevalence of Brugada syndrome varies from 5 to 20 cases in every 10,000 inhabitants worldwide, and the syndrome is considered endemic in Asian countries[3].

To date, the mechanisms underlying the syndrome remain to be clarified. Several hypotheses have been proposed involving abnormalities in both repolarization and depolarization; however, the same mechanism might not be responsible for the disease in all patients, and several might coexist in a single patient. Indeed, the Brugada syndrome phenotype might be caused by numerous different pathophysiological mechanisms.

Unlike other inherited primary arrhythmia syndromes, SCD can be the first manifestation of the disease in patients with Brugada syndrome[7]. Accurate risk stratification has a central role in the management of these patients. Patients at the highest risk for SCD are easily identified through clinical and electrocardiogram (ECG) examinations, but those at lower risk are more difficult to stratify, and they can still present with arrhythmic events. Furthermore, SCD can occur several decades after the initial diagnosis. Given that Brugada syndrome is an inherited condition, diagnosis in one individual should be followed by exhaustive familial screening; SCD might be present in several individuals of the same family and familial antecedents of SCD might have prognostic value.

Genetic basis

In the past 6 years, numerous pathogenic mutations have been found in several genes that encode for subunits of cardiac sodium, potassium, and calcium channels, as well as genes involved in the regulation of these channels (including *SCN5A*, *SCN10A*, *SCN1B*, *SCN2B*, *SCN3B*, *GPD1L*, *RANGRF*, *SLMAP*, *KCNE3*, *KCNJ8*, *KCNE5*, *KCND3*, *HCN4*, *CACNA1C*, *CACNB2B*, *CACNA2D1*, *TRPM4*, and *PKP2*)[8–10]. However, many of these identified gene mutations are not thought to be directly responsible for the disease, at least not alone (TABLE 1).

Mutations in the cardiac sodium channel gene *SCN5A*, which encodes the α-subunit of the human cardiac sodium channel, have been identified in 11–28% of patients with Brugada syndrome[8,11,12]. *SCN5A* expression influences phase 0 of the cardiac action potential, and pathogenic variations result in sodium channel

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

Key points

- Brugada syndrome is an inherited disease responsible for a large number of sudden deaths in young people without structural heart anomalies
- The pathophysiology of Brugada syndrome is unclear; repolarization–depolarization abnormalities underlying this disease can present with different features in different patients
- A common electrocardiogram (ECG) phenotype seen in patients with Brugada syndrome might be the result of different pathophysiological mechanisms
- Symptom-type, a spontaneous type 1 ECG pattern, the presence of sinus node dysfunction, and inducible ventricular arrhythmias during programmed stimulation of the heart might help to stratify patients by risk
- Implantable cardioverter–defibrillator therapy is the most accepted treatment strategy for at-risk patients
- Electrophysiological-guided therapy with quinidine and radiofrequency substrate ablation might also have a role in the management of these patients

dysfunction. To date, >300 different mutations of *SCN5A* have been identified and the list is still growing[8,9,12]. Many of the other Brugada syndrome-associated genes have a role in regulating sodium channel function, and several pathogenic variations in *SCN1B*, *SCN2B*, and *SCN3B* that encode the β-subunits of the $Na_v1.5$ sodium channel have been shown to modify channel function. *SCN10A*, a gene encoding the neuronal sodium channel $Na_v1.8$, can modulate *SCN5A* expression and the electrical function of the heart[8,9]. Pathogenic variations in *GPD1L* have been shown to reduce both the surface membrane expression and the inward peak sodium current (I_{Na}) of $Na_v1.5$, whereas mutations in *RANGRF* have been reported to impair the trafficking of $Na_v1.5$ to the membrane, leading to I_{Na} reduction and clinical manifestation of Brugada syndrome[12].

In addition to mutations in sodium channels, patients with Brugada syndrome are reported to have genetic variations in voltage-gated potassium channels (*KCNE3*, *KCNJ8*, *KCND3*, and *KCNE*) and in voltage-dependent calcium channels (*CACNA1C*, *CACNB2B*, and *CACNA2D1*)[12]. Variants have also been found in *HCN4*, which encodes a nucleotide-gated potassium channel located within the sinus node and cells of the cardiac conduction system; loss-of-function pathogenic variants in *HCN4* are associated with sinus nodal dysfunction[13]. Furthermore, *TRPM4*, the gene encoding the transient receptor potential cation channel subfamily M member 4 protein (or hTRPM4), is a member of a large family of transient receptor potential genes involved in conduction blockade and Brugada syndrome[14,15]. Activation of hTRPM4 enables sodium entry into the cell, leading to cellular membrane depolarization. hTRPM4 also participates in intracellular calcium sensing and influences the driving force for calcium and other ions by altering the cellular membrane electrical potential. A reduction or increase in hTRPM4 channel function, owing to mutations in TRPM4, might reduce the availability of the sodium channel and lead to the development of Brugada syndrome[14,15].

In addition to these ion-channel genes, numerous other genes are thought to be potentially involved in the pathogenesis of Brugada syndrome, such as the *PKP2* gene encoding for the desmosomal protein plakophilin 2. Sato and colleagues were first to demonstrate that I_{Na} was significantly reduced in cardiomyocytes lacking *PKP2* (REFS 16,17). Moreover, the loss *PKP2* expression might be associated with the development of Brugada syndrome in affected patients who do not present with arrhythmogenic cardiomyopathy or have mutations in Brugada syndrome-related genes (*SCN5A*, *CACNA1C*, *GPD1L*, and *MOG1*)[18]. However, to date, there has been no evidence to indicate that this gene has a role in the pathogenesis of Brugada syndrome. In 2013, Bezzina *et al.* conducted a genome-wide association study to explore the link between common genetic variants and susceptibility to Brugada syndrome[19]. The association signals at *SCN5A* and *SCN10A* demonstrate that genetic polymorphisms modulating cardiac conduction can also influence susceptibility to cardiac arrhythmia. The researchers also identified the *HEY2* gene as a transcriptional regulator of cardiac electrical function that is involved in the pathogenesis of Brugada syndrome. This gene encodes a member of the hairy and enhancer of split-related family of basic helix–loop–helix (bHLH)-type transcription factors. Two similar and redundant genes in mice are required for embryonic cardiovascular development, which are also implicated in neurogenesis and somitogenesis[19]. However, further pathogenic mutations associated with the syndrome could be localized in unknown genes, and incomplete penetrance and variable expressivity of these mutations are likely. In up to 60% of patients with Brugada syndrome, no other family members are found to be affected[20]. Finally, most of the other gene defects identified in the context of Brugada syndrome are suspected not to be directly responsible for the disease, at least not alone. Several studies have demonstrated that rare coding variations in previously reported arrhythmia-susceptibility genes, with the exception of *SCN5A*, do not contribute substantially to the development of Brugada syndrome[21,22].

The use of genetic testing in cardiology practice has gained momentum in the past decade owing to advances in next-generation sequencing technology. Given the complexity and limitations of genetic testing, however, commercially available test kits or assays might serve only to complicate the diagnostic procedure for practising clinicians. The degree to which different mutations confer risk of arrhythmic events or sudden death remains unclear and, therefore, does not effectively contribute to risk stratification. To date, genetic screening is recommended only for family members of Brugada syndrome probands who have been successfully genotyped; genetic testing is useful in these individuals for advancing research, and for establishing genotype–phenotype relationships and identifying family members who might be at risk[23].

Cellular mechanisms

The cellular mechanisms underlying Brugada syndrome remain debatable. Two principal hypotheses have been proposed: the repolarization hypothesis and the depolarization hypothesis.

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

Table 1 | **Genes associated with Brugada syndrome**

Gene	Protein	Chromosomal locus	Proportion of Brugada syndrome attributed to genetic variants (%)
Calcium channel			
CACNA1C	Voltage-dependent L-type calcium channel subunit α1c	12p13.33	6.6
CACNB2B	Voltage-dependent L-type calcium channel subunit β2	10p12.33-p12.31	4.8
CACNA2D1	Voltage-dependent calcium channel subunit α2/δ1	7q21.11	1.8
TRPM4	Transient receptor potential cation channel subfamily M member 4	19q13.33	<1
Sodium channel			
SCN5A	Sodium channel protein type 5 subunit α	3p21	11–28
SCN10A	Sodium channel protein type 10 subunit α	3p22.2	5.0–16.7
GPD1L	Glycerol-3-phosphate dehydrogenase 1-like protein	3p22.3	<1
SCN1B	Sodium channel subunit β1	19q13.12	1.1
SCN2B	Sodium channel subunit β2	11q23.3	<1
SCN3B	Sodium channel subunit β3	11q24.1	<1
RANGRF	Ran guanine nucleotide release factor	17p13.1	<1
SLMAP	Sarcolemmal membrane-associated protein	3p14.3	<1
PKP2	Plakophilin-2	12p11	<1
Potassium channel			
KCND3	Potassium voltage-gated channel subfamily D member 3	1p13.2	<1
KCNE3	Potassium voltage-gated channel subfamily E member 3	11q13.4	<1
KCNJ8	ATP-sensitive inward rectifier potassium channel 8	12p12.1	2.0
HCN4	Potassium/sodium hyperpolarization-activated cyclic nucleotide-gated channel 4	15q24.1	<1
KCNE5	Potassium voltage-gated channel subfamily E regulatory β subunit 5	Xq22.3	<1

The repolarization hypothesis

The repolarization hypothesis suggests that an outward shift in the balance of currents in the right ventricular epicardium can result in repolarization abnormalities (FIG. 1). These abnormalities can facilitate the development of phase 2 re-entry, which generates closely coupled premature beats that can precipitate ventricular tachycardia or ventricular fibrillation[24–26]. According to this theory, a reduced inward sodium current and prominent outward current lead to the accentuation of the action potential notch in the right ventricular epicardium relative to the endocardium, producing a transmural voltage gradient which manifests electrocardiographically as the characteristic ST-segment elevation seen in patients with Brugada syndrome. At the end of phase 1, certain epicardial sites undergo all-or-none repolarization, losing their action potential dome and resulting in the development of a local epicardial dispersion of repolarization. This heterogeneous repolarization environment leads to phase 2 re-entry and coupled extrasystoles when action potential domes migrate from sites where they are present to sites where they were lost[11]. Szel and colleagues performed a study to test this hypothesis directly[27]. Their findings indicated an alternative mechanism in which low-voltage fractionated electrogram activity develops owing to regional desynchronization in the second action potential upstroke, secondary to accentuation of the epicardial action potential notch. Furthermore, high-frequency late potentials developed in the right ventricular epicardium secondary to concealed phase 2 re-entry. Delayed conduction of the primary beat has not been observed in preclinical models of Brugada syndrome, created by exposing canine right ventricular wedge preparations to drugs designed to mimic the different genetic defects known to give rise to the disease[27].

The similarity between Brugada syndrome and early repolarization syndrome (ERS), with respect to clinical manifestations and the response to autonomic influences and therapy, lends additional support to the repolarization hypothesis. Using a canine left ventricle model of ERS, Koncz and colleagues demonstrated that accentuation of transmural gradients in the left ventricular wall is responsible for the repolarization abnormalities underlying ERS, much like in Brugada syndrome, giving rise to J-onset elevation, distinct J waves, or slurring of the terminal part of the QRS[28].

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

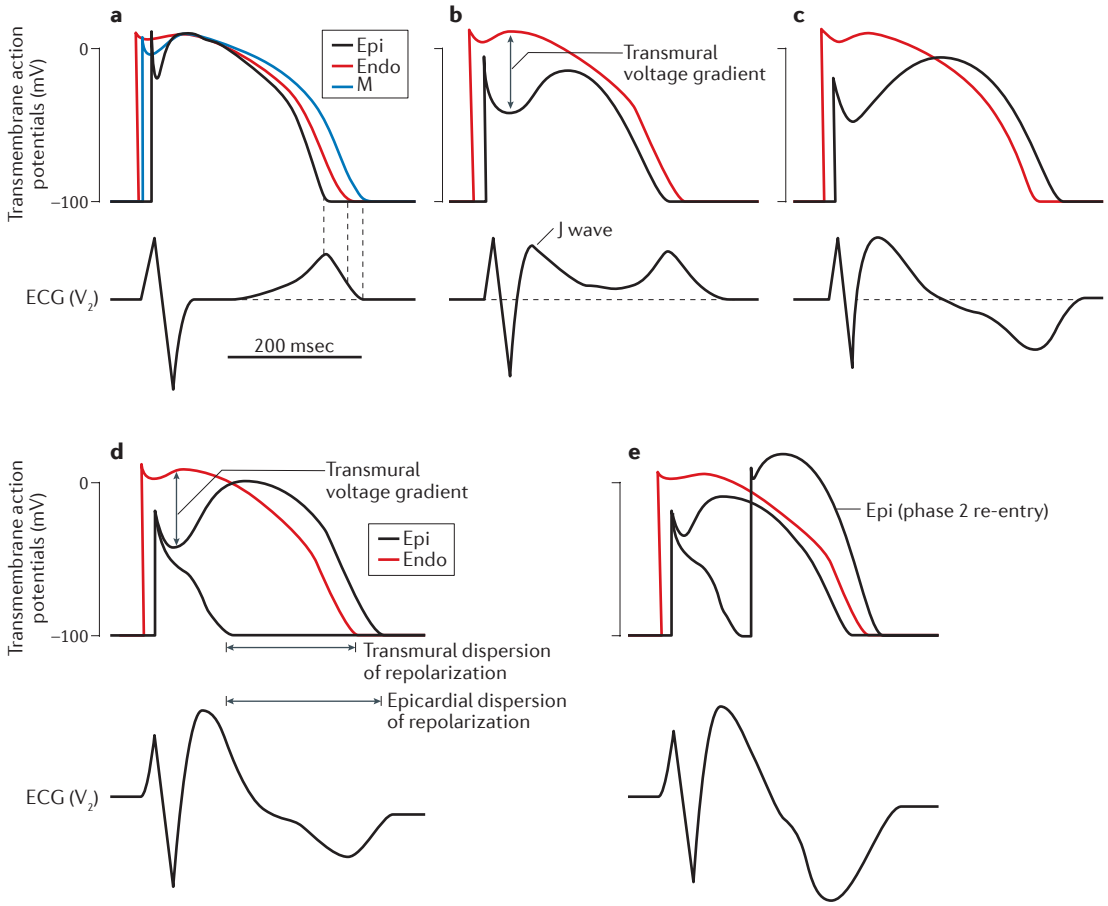

Figure 1 | **The repolarization theory of Brugada syndrome.** An outward shift in the balance of currents in the right ventricular epicardium can result in repolarization abnormalities seen in Brugada syndrome. **a** | Healthy individual. **b** | Early repolarization syndrome. **c** | Brugada syndrome (saddleback). **d** | Brugada syndrome (coved). **e** | Brugada syndrome (heterogenous loss of action potential dome). **f** | Brugada syndrome (phase 2 re-entry). ECG, electrocardiogram; Endo, endocardium; Epi, epicardium; M, midmyocardium. Reprinted from Antzelevitch, C. The Brugada syndrome: ionic basis and arrhythmia mechanisms. *J. Cardiovasc. Electrophysiol.* **12**, 268–272 (2001), with permission from John Wiley and Sons.

Furthermore, Patocskai and Antzelevitch showed that ablation of right ventricular epicardial tissue diminishes the appearance of J waves and ST-segment elevation, and abolishes all arrhythmic activity by destroying the cells with the most prominent action potential notch[29]. Therefore, the cells responsible for the repolarization abnormalities that give rise to phase 2 re-entry and ventricular tachycardia or fibrillation are ultimately eliminated[29].

The depolarization hypothesis

The depolarization hypothesis claims that slow conduction in the right ventricular outflow tract (RVOT) has a primary role in the development of the electrocardiographic and arrhythmic manifestations of Brugada syndrome[30]. According to this hypothesis, ST-segment elevation is caused by a conduction delay in the RVOT, and ventricular arrhythmias associated with Brugada syndrome are induced by the abnormal current created by delayed depolarization of the RVOT (FIG. 2). This model is based on the mechanism underlying ST-segment elevation in regional transmural ischaemia, in which a large potential difference between ischaemic and nonischaemic regions creates a current that presents as ST-segment elevation. The delayed depolarization of the RVOT with respect to the other right ventricular action potentials creates a potential difference between the right ventricle and the RVOT[11].

Findings from electrophysiological investigations with endocardial and epicardial mapping and ablation of patients with Brugada syndrome and electrical storm favour the depolarization theory by showing extremely slow conduction of the electrical impulse at the RVOT region. By far the most compelling evidence in support of the depolarization hypothesis was derived from findings by Nademanee and colleagues, who showed that radiofrequency ablation of epicardial sites displaying late potentials and fractionated bipolar electrograms in the RVOT of patients with Brugada syndrome significantly reduced arrhythmia vulnerability and the degree of ST-segment elevation[24]. The investigators concluded that the late potential and fractionated electrogram activity are attributable to conduction delays within the RVOT, and that ablation of the presumed sites of slow conduction is the basis for the ameliorative effect of ablation therapy[31].

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

The depolarization hypothesis has also been assessed in the context of the murine embryonic outflow tract, a slow conducting structure with low expression levels of connexin 43 (Cx43, also known as gap junction α1 protein), connexin 40 (also known as gap junction α5 protein), and *Scn5a* (REF. 32). Low expression of *Scn5a* and Cx43 reduces the safety of propagation (also known as conduction reserve) and increases susceptibility to arrhythmias. Boukens *et al.* hypothesized that the adult RVOT myocardium retains aspects of this embryonic outflow tract phenotype, thereby contributing to lower conduction reserve in the RVOT[32]. The embryonic RVOT is marked by the expression of *Tbx2*, and absence of *Gja1* and *Hey2*. To investigate whether this embryonic signature is maintained in the RVOT, the researchers measured the expression patterns of these markers in early fetal and late fetal mouse hearts by *in situ* hybridization. Furthermore, they assessed the functional consequences of reduced sodium current on RVOT conduction by pharmacological sodium-channel blockade in a mouse model with a human cardiac sodium-channel mutation associated with Brugada syndrome. A *Tbx2*-positive and *Gja1*-negative myocardial gene programme, and the slow-conducting phenotype of the embryonic outflow tract myocardium, was shown to be maintained in the RVOT until birth. In the adult heart, lower expression levels of *Gja1* and *Scn5a* were observed in the RVOT, and were associated with lower conduction reserve in the RVOT than in the right ventricle, resulting in more pronounced conduction slowing in the RVOT when sodium current was decreased. These findings provide an explanation as to why conduction delays occur preferentially in the RVOT, as observed in patients with Brugada syndrome[33,34].

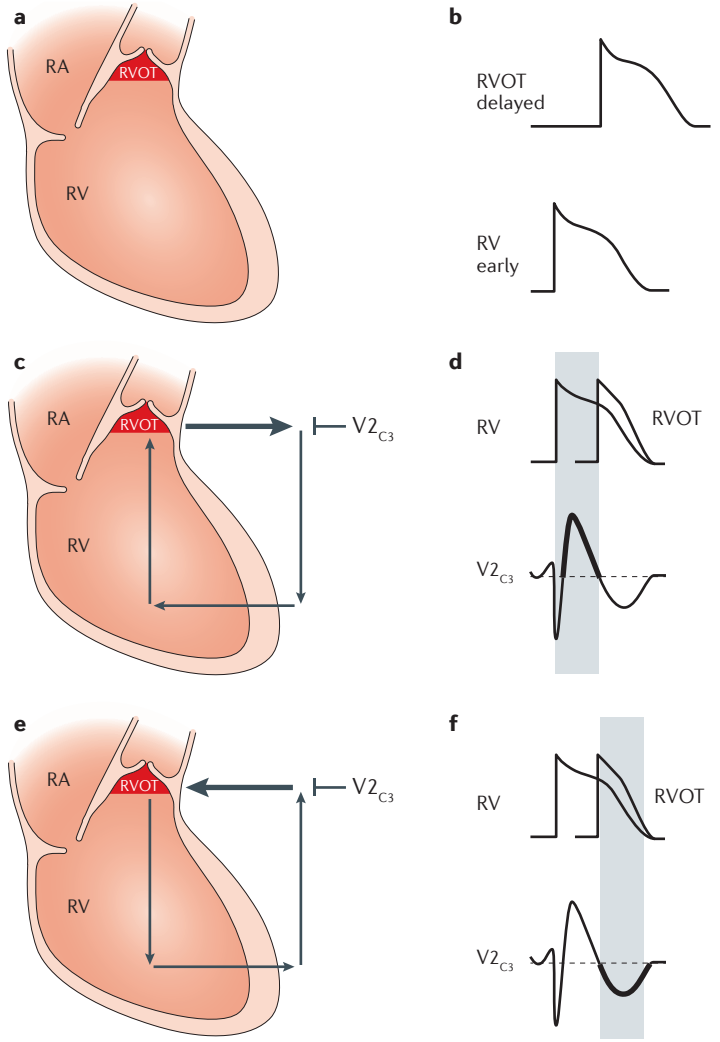

Figure 2 | **The depolarization theory of Brugada syndrome.** Slow conduction in the right ventricular outflow tract (RVOT) might have a role in the development of the electrocardiographic and arrhythmic manifestations of Brugada syndrome. **a** | Right side of the heart. **b** | The action potential of the RVOT is delayed with respect to the action potential of the right ventricle (RV). Panels **c** and **d** show the current flowing from the RV to the RVOT and into the extracellular space during the hatched phase of the cardiac cycle. Panels **e** and **f** show a reversal of the potential gradients between RV and RVOT, with the current passing away from lead $V2_{IC3}$. RA, right atrium. Reprinted from Meregalli, E. G. *et al.* Pathophysiological mechanisms of Brugada syndrome: depolarization disorder, repolarization disorder, or more? *Cardiovasc. Res.* **67**, 367–378 (2005), with permission from Oxford University Press.

The neural crest hypothesis

The repolarization and depolarization mechanisms underlying the development of Brugada syndrome are not necessarily mutually exclusive, and might indeed have a synergistic role[35]. In an attempt to convey a single explanation for the repolarization and the depolarization hypothesis, Elizari *et al.* proposed that the RVOT and its nearby structures have different embryological origins compared with the rest of the heart, and consequently possess different physiological, anatomical, and clinical characteristics[36]. Brugada syndrome might be linked to an abnormal expression of the neural crest during embryological development of the RVOT and surrounding structures (FIG. 3). According to this hypothesis, abnormal expression of the cardiac neural crest cells would lead to abnormal connexin expression, particularly Cx43, thus creating a depolarization delay and heterogeneity in repolarization of the RVOT necessary to induce the ECG characteristics observed in patients with Brugada syndrome[36].

Connexins, particularly Cx43, are involved in neural crest cell migration and cardiac impulse propagation. Faster and slower migration of the cardiac neural crest have been directly correlated with overexpression and underexpression of Cx43, respectively[37–39]. The degree of Cx43 function might be of critical importance in downstream events involving migration of neural crest cells and heart defects in the right ventricle[38,40]. Abnormal myocardialization linked with cardiac neural crest expression in the RVOT might explain repolarization–depolarization heterogeneities underlying the Brugada syndrome phenotype. In this context, improper gap junctional communication in the RVOT could lead to errors in cardiac neural crest cell expression, which can result in tissue remodelling and altered gap junctional channel configuration[37,40]. Transmural and regional heterogenic Cx43 distributions resulting from tissue remodelling can lead to slowed conduction and delayed action of the RVOT, mechanisms that underlie Brugada syndrome[36]. In 2000, Dhar Malhotra and colleagues demonstrated

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

that sodium channel β-subunits can also mediate cell adhesion, which has implications not only for the formation of the sodium channel complex, but also for sodium channel-independent functions such as cell migration, cell aggregation, and interaction with the cytoskeleton[41,42]. Moreover, several studies have confirmed and expanded the importance of Cx43 in the control of the sodium current amplitude[43–47]. Therefore, to date, the molecules necessary for making the gap junctions are hypothesized also to be necessary for maintaining the complex in charge of generating the action potential. Cx43-dependent arrhythmias can develop via mechanisms that are unrelated to impairment of gap junction channel function[41,45].

Concerning the role of Cx43 in Brugada syndrome, a series of six postmortem cases revealed that fibrosis was increased in the hearts of patients with Brugada syndrome compared with control hearts, particularly in the RVOT and in the epicardium; however, Cx43 expression was reduced in the RVOT[48]. Both the autopsy report and the *in vivo* RVOT samples showed evidence of epicardial and interstitial fibrosis, and reduced gap junction expression correlated with abnormal action potentials. However, upon ablation, the Brugada syndrome phenotype and associated ventricular arrhythmias were abolished[48]. Given that no genetic variants have been found in *CX43* as yet, the hypothesis that genetic abnormalities in *CX43* or in other related genes contribute to the pathogenesis of Brugada syndrome requires further investigation.

In their review, Agullo-Pasqual and colleagues hypothesized that molecules classically defined as belonging to one particular structure as desmosomes, gap junctions, or sodium channel complexes actually interact with one another and with other cellular structures in a common protein interacting network called 'the connexome' to control excitability, electrical coupling, and intercellular adhesion in the heart[49]. The perinexus was identified as a microdomain surrounding the cardiac gap junctions that contains elevated levels of Cx43 and the sodium channel protein $Na_v1.5$. The perinexus is a specialized domain of the plasma membrane that is enriched with connexons and patterns of protein–protein interactions that can provide opportunity for regulation of hemichannel activity. The myocyte perinexus contains high concentrations of connexons, which suggests that the function of the Cx43 perinexus might be to provide the framework for an ephapse between cardiomyocytes[50,51]. Ongoing studies have established a role for the perinexus in regulating gap junction aggregation[52]; from this perspective, this cell structure could be the site of the connexome[49].

Desmosomal mutations are known to cause arrhythmogenic cardiomyopathy, and sodium channel dysfunction is involved in Brugada syndrome; however, if the molecular substrates are indeed part of a common network, these two diseases should also share some common features. Though the two syndromes are distinct clinical entities, they represent 'bookends' of a spectrum of clinical manifestations that vary depending on the effect that a particular mutation has on the connexome as a whole. Therefore, these diseases might share a common origin as diseases of the connexome[49].

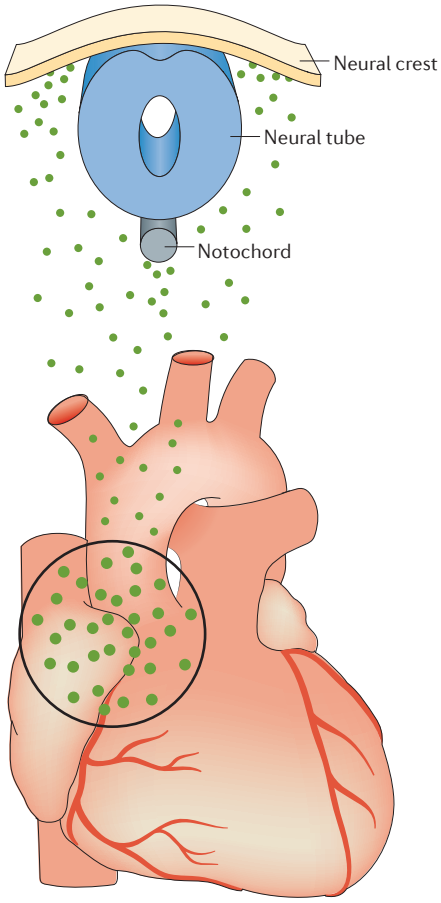

Figure 3 | **Neural crest cell migration.** Cardiac neural crest cells (green dots) migrate from the neural crest to the right ventricular outflow tract, large vessels, and surrounding structures. Brugada syndrome might be linked to an abnormal expression of the neural crest during embryological development of these structures. Reprinted from Elizari, M. V. *et al.* Abnormal expression of cardiac neural crest cells in heart development: a different hypothesis for the etiopathogenesis of Brugada syndrome. *Heart Rhythm* **4**, 359–365 (2007), with permission from Elsevier.

To date, no single clinical or experimental study has provided irrefutable proof in support of a lone hypothesis regarding the pathophysiology of Brugada syndrome while rejecting all other hypotheses. The wide spectrum of the severity of the electrocardiographical and clinical manifestations of Brugada syndrome correlates with the existence of a wide array of underlying genetic and cellular abnormalities of the RVOT and surrounding structures. This observation begs the question: can the pathogenesis of Brugada syndrome be explained using a single mechanism, or are we dealing with different diseases with a common phenotype? The repolarization–depolarization abnormalities underlying the Brugada syndrome phenotype (whether they be purely related to genetic alterations, or owing to abnormal neural crest cell migration during myocardial development of the RVOT, improper gap junctional communication, or to connexome abnormalities), are all part of a complex polyfactorial pathological system that cannot be simplified by a single

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

causal mechanism. Therefore, the manifestation of the typical Brugada ECG pattern might be attributable to all these mechanisms equally or the preponderance of one over another, and are not necessarily mutually exclusive[11].

As in long QT syndrome or hypertrophic cardiomyopathy, Brugada syndrome might be an accumulation of different diseases with a common phenotype. For example, mutations in the potassium channels lead to forms of long QT syndromes that are pathophysiologically unrelated to long QT syndrome type 3, which is caused by mutations in the cardiac sodium channel[53,54]. In hypertrophic cardiomyopathy, the hypertrophic phenotype is the common link of all the different variants[55]. However, the distinct course and pathophysiological mechanisms of the different genetic variants clearly show that they are distinct diseases with a common phenotype. The same can be said for Brugada syndrome: a common clinical phenotype (ECG abnormalities) can be caused by different pathophysiological mechanisms depending on the type and location of the mutations[11,56].

Management

Management of Brugada syndrome remains a challenging and controversial issue. Prompt recognition of patients with Brugada syndrome who are at risk of arrhythmic events is critical. Patients at the highest risk are easily identified as they might present after an aborted SCD or with very expressive clinical and ECG features. However, patients at low risk of arrhythmic events are much harder to recognize.

In the >20 years since the first description of Brugada syndrome, the clinical characterization of patients has evolved considerably[1,57]. The early reports of patients with Brugada syndrome described individuals with a very expressive form of the disease, who were frequently symptomatic and displayed the type 1 ECG pattern spontaneously (characterized by an ST-segment elevation of 2 mm [0.2 mV], followed by a negative T wave)[2]. Ventricular arrhythmias were also induced more frequently[1,57]. At present, Brugada syndrome is diagnosed more often in asymptomatic individuals or after a drug challenge test because of suspicion of the disease[58]. However, most Brugada syndrome registries and studies do not adequately represent this shift in patient characterization[59] and, therefore, management of this 'new' subset of patients is not well established. Furthermore, current clinical guidelines and consensus statements offer recommendations for those patients at higher risk[60,61], whereas asymptomatic and drug-induced patients with Brugada syndrome are almost never mentioned.

Several factors contribute to this shift in patient characteristics. First, clinicians might be more aware of Brugada syndrome and, therefore, clinical suspicion would be more easily raised. Second, clinicians now have a better knowledge of the disease, specifically its relationship with atrial fibrillation, and patients might be diagnosed when presenting with this comorbidity. Third, familial screening is becoming more widespread. A combination of these factors might explain the increasing number of asymptomatic individuals diagnosed with Brugada syndrome.

An implantable cardioverter–defibrillator (ICD) is recommended for the prevention of SCD in patients with Brugada syndrome[60]. However, given that Brugada syndrome is usually diagnosed in young and otherwise healthy individuals, the decision to implant an ICD requires thorough evaluation[58] owing to the risk of long-term device complications[62]. Furthermore, inappropriate shocks are likely in this population given their active lifestyle, which might have a negative psychological effect in the long term. In a meta-analysis published in 2016, 20% of patients with Brugada syndrome who received an ICD presented with inappropriate shocks, and 20% presented with device-related complications (12% of which were the result of electrode dysfunction)[62]. Similar findings have been reported in other studies[63,64]. These complications are relevant and should be taken into consideration during the assessment and management of patients with Brugada syndrome. Nevertheless, life-threatening complications are extremely infrequent. The development of completely extravascular devices that reduce the risk of systemic device-related infections, and novel therapies such as epicardial substrate ablation[31,65] and quinidine treatment[66] might improve the management of patients with Brugada syndrome.

Risk factors for arrhythmic events

The decision on whether to implant an ICD or conservatively manage patients with Brugada syndrome can be made only after careful evaluation of the risk of a predicted arrhythmic event. Overall, the annual arrhythmic incidence rate for patients with Brugada syndrome is between 1.3% and 1.6%[59,67]. However, the risk of events can be dramatically altered depending on clinical characteristics; a patient presenting after an aborted SCD has a 7.7% annual risk of a new cardiac arrest[59]. Conversely, an asymptomatic female patient with Brugada syndrome has an annual incidence rate of 0.25% for SCD[68], which is >100 times that of an age-matched, healthy female individual[69]. Several risk factors for SCD have been widely reported and are almost universally accepted, but others remain controversial.

Symptoms. The symptoms presented by a patient with Brugada syndrome at diagnosis are one of the most important risk markers for future arrhythmic events. Patients diagnosed after an aborted SCD are at the highest risk, with almost 60% of patients presenting with a new event 10 years after the initial diagnosis[58,63]. Syncope is also a strong predictor of events, and patients who experienced syncope have an arrhythmic event rate of 1.9% per year and hazard ratio of 3.6 compared with asymptomatic individuals[58]. The combination of syncope and a spontaneous type 1 ECG pattern is associated with poor prognosis[70]. Current guidelines recommend ICD implantation in these patients; however, several studies have demonstrated that both factors are independent and should be considered individually[59,71].

The subgroup of patients with Brugada syndrome who are asymptomatic require careful management. These patients are theoretically at a lower risk than those who present with symptoms, but are also susceptible to

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

arrhythmic events during follow-up, even in the absence of previous symptoms or other risk factors[6]. Interestingly, the event rate of asymptomatic patients who received an ICD was reported to be similar to that of patients with a history of syncope[63]. Approximately two-thirds of patients in our cohort of patients with Brugada syndrome were asymptomatic[58,72]. Similar figures have also been reported by other large registries: 63% in the FINGER registry[1] and 78% in the PRELUDE registry[11]. As mentioned previously, the clinical profile of patients with Brugada syndrome at presentation has evolved over time. Whereas earlier reports described patients with a more expressive clinical and ECG phenotype, at present, patients are frequently asymptomatic with a nonspontaneous type 1 pattern at diagnosis[58]. In 2002, an annual event rate of 2.7% was reported in asymptomatic patients[73], but this figure has since fallen to 0.5%, as reported by more recent registries; this discrepancy is likely attributable to selection bias[59,72]. Although this event rate might seem relatively low compared with symptomatic patients, it remains extremely relevant given the young age and the long life expectancy of these patients who have no other pathological conditions. The risk of SCD in a healthy individual aged 40 years is approximately 100 times less than in asymptomatic patients with Brugada syndrome[69,74]. Nevertheless, in patients with a type 1 Brugada ECG pattern, ICD-recorded ventricular arrhythmias are not a perfect surrogate measure of SCD and might result in overestimation of event rate[75].

Age. The age of an individual at diagnosis of Brugada syndrome is unrelated to prognosis[63,64]. Patients are typically diagnosed aged 40–50 years[72]. Two subgroups of patients merit special consideration: paediatric and elderly patients. Fortunately, the prevalence of Brugada syndrome in patients aged <18 years is low. Nevertheless, among the eight patients that constituted the initial report on Brugada syndrome, three were children[1]. Paediatric patients with Brugada syndrome who present with symptoms have particularly poor prognosis, with an event rate of 4.5% per year[76,77]. Conversely, asymptomatic patients seem to have good outcomes, particularly when they do not show a spontaneous type 1 Brugada pattern on ECG[78]. Individual evaluation is required, and patients deemed to be at high risk of events should undergo ICD implantation.

Brugada syndrome is uncommonly diagnosed in elderly patients and is associated with a benign prognosis[79]. Among all patients with Brugada syndrome, those aged >70 years did not present with arrhythmic events during follow-up[79]. Nevertheless, establishing the diagnosis of this syndrome in elderly patients remains important, given the familial pattern of inheritance, and the implications on diagnosis of other family members.

Sex. Brugada syndrome was thought to be a condition affecting mainly men[59,70], but a study published in 2016 reported that women comprised 42% of a cohort of 542 patients who presented with a spontaneous or drug-induced Brugada type 1 pattern on ECG[68]. Sex might be associated with prognosis in patients with Brugada syndrome. Compared with their male counterparts, women with Brugada syndrome present with more benign clinical characteristics, less spontaneous type 1 ECG pattern, and are more likely to be asymptomatic[80]. Nevertheless, arrhythmic events can occur during follow-up[81] and, importantly, clinicians currently lack specific risk factors to stratify risk based on sex[68]. Several studies have attempted to address this issue. Benito and colleagues reported that a history of atrial fibrillation and a prolonged PR interval were associated with increased risk of arrhythmic events in women with Brugada syndrome[80]. Furthermore, we reported that a history of sinus node dysfunction (SND) was related to poor prognosis. Interestingly, a spontaneous type 1 ECG pattern is not associated with increased risk of events in this subgroup of patients[68].

Familial history of SCD. Notably, a family history of SCD is not associated with worse prognosis in patients with Brugada syndrome[72]. Our group has reported that multiple antecedents of SCD in first-degree relatives aged <35 years were associated with more arrhythmic events, but this condition was uncommon, and the association was no longer significant after adjusting for other variables[82].

Specific gene mutations. As previously mentioned, much effort has been made in the identification of genetic mutations related to the pathophysiology of Brugada syndrome. Conte and colleagues reported that mutations can be found in approximately 20–30% of patients with Brugada syndrome[63], and others have reported an even higher proportion[76,83]. Whether the presence of genetic variants can result in increased arrhythmic risk and poor prognosis remains a matter of debate[59,84]. In a study involving patients carrying loss-of-function *SCN5A* mutations linked to Brugada syndrome, differential mutation patterns were associated with different degrees of disease severity[85].

ECG parameters. Given that Brugada syndrome is a disease affecting electrophysiological activity, ECG patterns merit special consideration for both diagnostic and prognostic issues. Recognition of the typical ECG pattern is of utmost importance for risk stratification of patients with Brugada syndrome. Current clinical guidelines and consensus documents state that Brugada syndrome is diagnosed in patients with ST-segment elevation with type 1 morphology ≥2 mm in one or more leads among the right precordial leads V1 and/or V2 positioned in the second, third, or fourth intercostal space, occurring either spontaneously or after provocative drug testing with intravenous administration of sodium-channel blockers (such as ajmaline, flecainide, pilsicainide, or procainamide)[60,61]. Several issues in these new guidelines should be highlighted. First, the ECG pattern in only one lead should be sufficient for the diagnosis, because these patients have been shown to have a similar prognosis to those displaying the Brugada ECG pattern in more than one lead[86]. Second, the pattern can be also be demonstrated after a drug challenge test. Finally, higher intercostal leads are of equal diagnostic and prognostic value compared with the standard leads[87].

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

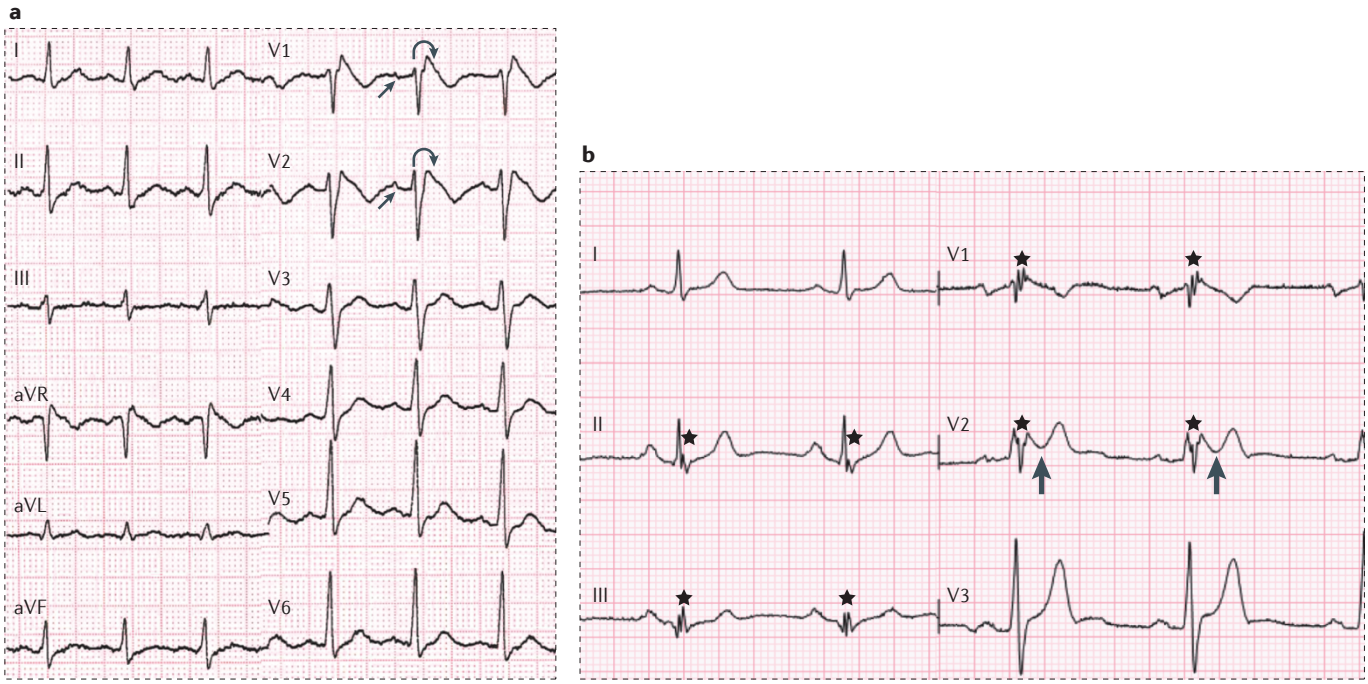

Figure 4 | **Electrocardiogram patterns associated with Brugada syndrome. a** | Spontaneous type 1 Brugada pattern. The straight arrow points to PR prolongation, whereas the curved arrow shows type 1 ST-segment elevation. **b** | Type 2 Brugada pattern with QRS fragmentation. The stars indicate the sites of QRS fragmentation, and the arrows indicate type 2 ST-segment elevation.

Brugada phenocopies — ECG patterns that are identical to true Brugada syndrome, but are induced by various clinical conditions — are also relevant when diagnosing and risk stratifying patients. In some cases, Brugada phenocopies can be extremely difficult to differentiate from true Brugada syndrome[88]. Therefore, clinical conditions that can produce these patterns should be carefully ruled out when evaluating a patient suspected of having Brugada syndrome.

The importance of the ECG pattern is not limited to the presence of the type 1 ECG pattern, and much effort has been made to identify other ECG characteristics that might help in risk stratification (FIG. 4). Several studies have described novel characteristics, such fragmentation of the QRS complex[89–91]. However, clarifying the value of these novel characteristics has proven to be difficult, and given that these novel ECG parameters have been described only in patients with a high-risk profile, the information they offer might be redundant.

The hallmark of Brugada syndrome is the characteristic ST-segment elevation. The ECG pattern offers important prognostic data, with patients who display the spontaneous type 1 pattern having a worse prognosis, with a hazard ratio of 4.0 for arrhythmic events as compared with those without the spontaneous pattern[72]. The typical ECG pattern of patients with Brugada syndrome might show spontaneous variations in both morphology and ST-segment elevation[2,92]. In addition to spontaneous fluctuation, many factors and drugs influence ST-segment elevation[93,94]. Approximately 20% of patients initially diagnosed after a drug challenge display the type 1 spontaneous pattern during follow-up[63].

Owing to their low-risk profile and the potential long-term complications of an ICD, risk stratification in the drug-induced Brugada syndrome subgroup needs to be very precise. Several reports have questioned the value of drug-induced diagnosis of Brugada syndrome as they suggest that the arrhythmic risk of these patients is very low[95,96]. However, to date, drug challenge remains the best available tool for diagnosis[97], given that the diagnostic value of the presence or absence of certain genetic variants requires further elucidation. Furthermore, within the same family, individuals with the same mutation can exhibit different responses during the drug challenge.

Several ECG parameters are reported to have prognostic value. The presence of QRS fragmentation on ECG has been identified in 30–40% of patients, and is associated with a worse prognosis[70,90,98]. The presence of QRS fragmentation is also linked to a more symptomatic disease phenotype. QRS fragmentation was present in approximately one-third of asymptomatic patients with Brugada syndrome in a study involving 115 patients, but none had experienced arrhythmic events during follow-up[98]. Another report, however, has proposed its role as an independent risk marker[99].

Together with QRS fragmentation, repolarization anomalies (so-called 'early repolarization') can also contribute to risk stratification. Early repolarization is present in approximately 10% of patients with Brugada syndrome, and might coexist with fragmented QRS[100]. An early repolarization ECG pattern is associated with a more severe clinical presentation and can also have independent prognostic value. In approximately 16% of patients with idiopathic ventricular fibrillation displaying

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

early repolarization in the inferior leads, a type 1 Brugada ECG pattern could be recorded in high intercostal leads[101]. Furthermore, a wide or prominent S-wave in lead 1 has been suggested to portend a worse prognosis in patients with Brugada syndrome[89]. However, this finding has been shown only in a study involving mostly asymptomatic patients with a spontaneous type 1 ECG pattern, and its value in the remaining Brugada syndrome population has not yet been proven[89]. Other ECG parameters that might be associated with a worse prognosis are the T-peak T-end interval, T-wave alternans, and the aVR sign (a prominent R in lead aVR)[102–105]. Evidence regarding their prognostic value is mainly derived from studies involving small numbers of patients and, therefore, their usefulness as an isolated risk marker remains to be confirmed.

Atrial fibrillation. Atrial fibrillation is more common in patients with Brugada syndrome than in the general population[106]. Furthermore, the presence of atrial fibrillation can raise clinical suspicion, and can often result in the diagnosis of the syndrome in a young patient[107]. The comorbidity of atrial fibrillation and Brugada syndrome is associated with a higher risk of events, more expressive clinical presentation, and worse long-term outcomes[108]. As with other parameters, the presence of atrial fibrillation can just be a marker of a more severe phenotype of the disease and not independently associated with prognosis.

Sinus node dysfunction. Given that the pathogenesis of SND is sometimes associated with mutations in the sodium-channel proteins, SND can be present in patients with Brugada syndrome[109,110]. Although the underlying mechanisms of SND in these patients are not clear, they usually present earlier and with a more severe phenotype, and are frequently symptomatic[72,111]. Nevertheless, asymptomatic patients with Brugada syndrome and concomitant SND have been reported to have worse prognosis than those with Brugada syndrome alone, justifying a more aggressive therapeutic strategy[72].

Table 2 | **Predictive value of EPS in Brugada syndrome**

Study	Number of patients	EPS predictive of events?	Inducibility rate (%)	Stimulation protocol
Eckardt *et al.* (2002)[113]	41	No	68	2 sites
Priori *et al.* (2002)[57]	86	No	66	Heterogeneous
Eckardt *et al.* (2005)[118]	186	No	50	2 sites
Takagi *et al.* (2007)[119]	146	No	73	2 sites
Probst *et al.* (2010)[59]	638	No	41	2 sites
Priori *et al.* (2012)[70]	308	No	40	2 sites
Brugada *et al.* (2001)[112]	252	Yes	51	1 site
Brugada *et al.* (2003)[120]	408	Yes	40	1 site
Sieira *et al.* (2015)[58]	403	Yes	18.1	1 site
Sroubek *et al.* (2016)[67]	1,312	Yes	40.2	Heterogeneous

EPS, electrophysiological study.

Programmed stimulation of the heart. The prognostic value of electrophysiological testing in Brugada syndrome is a controversial topic (TABLE 2). The earliest data supporting its use for the identification of high-risk patients were reported 15 years ago[112]. Since then, several studies have reported contradictory results[71,113]. In 2010, the FINGER registry[59], which included 1,029 patients with Brugada syndrome with spontaneous type 1 ECG patterns (median follow-up period of 32 months) reported that inducible ventricular arrhythmias had a borderline association with events in a univariate analysis. The subsequent PRELUDE registry[70], which was specifically designed to evaluate the role of electrophysiological testing in Brugada syndrome in 308 individuals (mean follow-up period of 34 months), reported that Kaplan–Meier event curves were practically identical in patients with or without inducible ventricular arrhythmia. In a subsequent study published in 2015 involving 404 patients with Brugada syndrome (median follow-up period of 74 months), electrophysiological testing was independently associated with a worse outcome (HR 8.3)[38]. Interestingly, electrophysiological testing was linked with a very high negative predictive value (98%), suggesting that patients without inducible ventricular arrhythmia have excellent prognosis, which might be particularly important in asymptomatic patients. In our experience, among 273 asymptomatic patients who underwent an electrophysiological study, only one noninducible patient presented with an event; the patient was male, had a drug-induced Brugada ECG pattern, familial antecedents of SCD, and no other risk factors[8]. Therefore, electrophysiological testing might be useful for identifying patients at high risk, and act as reassurance for those with a negative test.

A pooled analysis of electrophysiological studies in patients with Brugada syndrome was published in 2016, and consisted of data from eight registries with a total of 1,312 patients (median follow-up period of 38 months)[67]. The study concluded that electrophysiological studies could predict future arrhythmic events (HR 2.7), and importantly, inducibility was significantly associated with events when adjusted for variables, including ECG pattern and symptoms[67]. However, this pooled analysis did not include the studies with data that supported the prognostic value of electrophysiological studies.

The reasons underlying the contradictory results between studies remain unclear, but might be related to a multitude of factors. First, electrophysiological study protocols vary widely between different centres. Our protocol has remained unchanged since the first reports of our experience[58,112]. Our procedure involves only one stimulation site (the right ventricle apex), and it does not include repetition of extrastimuli. Compared with other study protocols, our protocol might be less aggressive, and the fact that our inducibility rate is one of the lowest published in the literature is a reflection of this point of difference. A more aggressive protocol might decrease the specificity of the test and might explain the divergent results found in the literature. The PRELUDE registry[70] used two stimulation sites,

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

whereas the FINGER registry[59] and the pooled analysis[67] published in 2016 do not have a homogenous protocol. Importantly, clinicians must remember that an electrophysiological study is not a diagnostic test, but rather a tool to help to risk stratify patients. An inducible ventricular arrhythmia does not necessarily mean that the patient will present with arrhythmic events, only that they are at higher risk.

General recommendations

Strategies to improve risk stratification in patients with Brugada syndrome are under investigation. In addition to classic risk factors, such as spontaneous type 1 ECG or symptoms, ongoing studies continue to report new risk factors that can aid in disease management. Unfortunately, as previously shown, these new markers are found predominantly in patients with more severe clinical presentation and, therefore, already easily recognized as being at high risk of events. The role of these new markers as independent risk factors is yet to be demonstrated.

Certain general measures should always be recommended to patients with Brugada syndrome. Aggressive treatment of fever and avoidance of particular drugs, especially those that block sodium channels, are important[60]. In addition, some considerations should be made regarding sports practice, although events related to Brugada syndrome are thought typically to occur during sleep or during vagal states, and events during exercise are uncommon[114]. However, arrhythmic events might occur immediately after exercise owing to a combination of increased vagal tone and hyperthermia. Exercise testing might have a role in risk stratification in certain patients[115].

An ICD is the most accepted therapy for high-risk patients. Clinical guidelines recommend an ICD in patients who have survived a SCD event (class Ia), patients with syncope and a spontaneous type 1 ECG pattern (class IIa), and those with inducible arrhythmias (class IIb)[60,61]. In our experience, four variables should be taken into consideration when evaluating the arrhythmic risk of a patient: symptoms, a spontaneous type 1 ECG pattern, the presence of SND, and inducible ventricular arrhythmias during programmed stimulation of the heart. Those with Brugada syndrome presenting after SCD are at extremely high risk of recurrent arrhythmias and should undergo ICD implantation. Patients with syncope are also at risk, particularly when presenting with a spontaneous type 1 ECG pattern. The current guideline recommendations for patients who experienced syncope were based on a study published in 2002 in which the syncope group consisted of only 34 patients with Brugada syndrome[57]. Given that several studies have since found that symptoms and the ECG pattern are independent risk factors[59,67], a more aggressive treatment strategy is justified. In addition, inducible ventricular arrhythmias have also been independently associated with events[67]. Lastly, the presence of SND should also be assessed with caution. Although uncommon, a patient with concomitant SND and Brugada syndrome might be at higher risk of events[72].

ECG findings are also important. Nevertheless, a spontaneous type 1 ECG pattern on its own does not justify ICD implantation[60]. Each patient should be evaluated individually, paying special consideration to risk factors such as sex, other ECG characteristics, and family history of SCD. In this context, a noninducible electrophysiological study is reassuring, given that the negative predictive value of the test is 0.98.

Non-device-based therapeutic tools

Quinidine is widely accepted as a treatment for electrical storm or frequent ICD shocks in patients with Brugada syndrome[60], or as an alternative for patients contraindicated for ICD implantation. Quinidine has been shown to be effective as an alternative to ICD, even in high-risk patients[66,116]. The use of quinidine in this setting involves a very aggressive electrophysiological stimulation protocol, with repetition of the test when taking the drug during follow-up. No arrhythmic events have been reported during follow-up in these patients[66]. However, other reports showed that quinidine does not completely suppress arrhythmic events in patients with Brugada syndrome[117]. Given that the mechanisms underlying the development of Brugada syndrome are multifaceted, and quinidine's actions are limited to inhibition of the transient outward potassium current, drug therapy does not guarantee complete protection. Further studies should be performed to clarify the role of this electrophysiological-guided drug therapy.

Epicardial radiofrequency substrate ablation has emerged as a promising tool for the management of Brugada syndrome. First described by Nademanee and colleagues in 2011, radiofrequency ablation of the anterior aspect of the RVOT rendered arrhythmias noninducible during electrophysiological testing and normalized ECG patterns, and patients had an excellent prognosis at 20 months[31]. Similar results have also been reported by others[65]. To date, this technique has been described only in small studies with short follow-up duration. Longer-term follow-up and multicentre studies are needed to clarify its efficacy and safety. Available data from highly experienced centres suggest a low complication rate[31,65].

Conclusions

The pathophysiology of Brugada syndrome remains under investigation. The typical ECG phenotype observed in patients with Brugada syndrome might involve a multitude of underlying mechanisms that can differ in each patient. A personalized approach to treatment and management of the disease might be the most effective strategy. Furthermore, identification of patients at risk of SCD remains the highest priority. Clinical and ECG characteristics are currently the most accurate and straightforward parameters for risk stratification in patients with Brugada syndrome. Novel markers are promising, but their added value as risk factors remains to be proved. ICDs are currently the most accepted therapy to protect patients at risk, particularly in those with a high-risk profile. Novel therapies such as electrophysiological-guided drug therapy or epicardial substrate ablation might be useful in selected patients.

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

1. Brugada, P. & Brugada, J. Right bundle branch block, persistent ST segment elevation and sudden cardiac death: a distinct clinical and electrocardiographic syndrome. A multicenter report. *J. Am. Coll. Cardiol.* **20**, 1391–1396 (1992).
2. Antzelevitch, C. *et al.* Brugada syndrome: report of the second consensus conference: endorsed by the Heart Rhythm Society and the European Heart Rhythm Association. *Circulation* **111**, 659–670 (2005).
3. Nademanee, K. *et al.* Arrhythmogenic marker for the sudden unexplained death syndrome in Thai men. *Circulation* **96**, 2595–2600 (1997).
4. Tan, H. L., Hofman, N., van Langen, I. M., van der Wal, A. C. & Wilde, A. A. Sudden unexplained death: heritability and diagnostic yield of cardiological and genetic examination in surviving relatives. *Circulation* **112**, 207–213 (2005).
5. van der Werf, C. *et al.* Diagnostic yield in sudden unexplained death and aborted cardiac arrest in the young: the experience of a tertiary referral center in The Netherlands. *Heart Rhythm* **7**, 1383–1389 (2010).
6. Mellor, G. *et al.* Clinical characteristics and circumstances of death in the sudden arrhythmic death syndrome. *Circ. Arrhythm. Electrophysiol.* **7**, 1078–1083 (2014).
7. Garson, A. Jr *et al.* The long QT syndrome in children. An international study of 287 patients. *Circulation* **87**, 1866–1872 (1993).
8. Kapplinger, J. D. *et al.* An international compendium of mutations in the SCN5A-encoded cardiac sodium channel in patients referred for Brugada syndrome genetic testing. *Heart Rhythm* **7**, 33–46 (2010).
9. Hu, D. *et al.* Mutations in SCN10A are responsible for a large fraction of cases of Brugada syndrome. *J. Am. Coll. Cardiol.* **64**, 66–79 (2014).
10. Nielsen, M. W., Holst, A. G., Olesen, S. P. & Olesen, M. S. The genetic component of Brugada syndrome. *Frontiers Physiol.* **4**, 179 (2013).
11. Dendramis, G., Antzelevitch, C. & Brugada, P. Brugada Syndrome: Diagnosis, Clinical Manifestations, Risk Stratification and Treatment. (Nova Science Publishers, 2015).
12. Sarquella-Brugada, G., Campuzano, O., Arbelo, E., Brugada, J. & Brugada, R. Brugada syndrome: clinical and genetic findings. *Genet. Med.* **18**, 3–12 (2016).
13. Ueda, K. *et al.* Role of HCN4 channel in preventing ventricular arrhythmia. *J. Hum. Genet.* **54**, 115–121 (2009).
14. Liu, H. *et al.* Molecular genetics and functional anomalies in a series of 248 Brugada cases with 11 mutations in the TRPM4 channel. *PLoS ONE* **8**, e54131 (2013).
15. Syam, N. *et al.* Variants of transient receptor potential melastatin member 4 in childhood atrioventricular block. *J. Am. Heart Assoc.* **5**, e001625 (2016).
16. Sato, P. Y. *et al.* Interactions between ankyrin-G, plakophilin-2, and connexin43 at the cardiac intercalated disc. *Circ. Res.* **109**, 193–201 (2011).
17. Sato, P. Y. *et al.* Loss of plakophilin-2 expression leads to decreased sodium current and slower conduction velocity in cultured cardiac myocytes. *Circ. Res.* **105**, 523–526 (2009).
18. Cerrone, M. *et al.* Missense mutations in plakophilin-2 cause sodium current deficit and associate with a Brugada syndrome phenotype. *Circulation* **129**, 1092–1103 (2014).
19. Bezzina, C. R. *et al.* Common variants at SCN5A-SCN10A and HEY2 are associated with Brugada syndrome, a rare disease with high risk of sudden cardiac death. *Nat. Genet.* **45**, 1044–1049 (2013).
20. Campuzano, O., Brugada, R. & Iglesias, A. Genetics of Brugada syndrome. *Curr. Opin. Cardiol.* **25**, 210–215 (2010).
21. Crotti, L. *et al.* Spectrum and prevalence of mutations involving BrS1- through BrS12-susceptibility genes in a cohort of unrelated patients referred for Brugada syndrome genetic testing: implications for genetic testing. *J. Am. Coll. Cardiol.* **60**, 1410–1418 (2012).
22. Le Scouarnec, S. *et al.* Testing the burden of rare variation in arrhythmia-susceptibility genes provides new insights into molecular diagnosis for Brugada syndrome. *Hum. Mol. Genet.* **24**, 2757–2763 (2015).
23. Ackerman, M. J. *et al.* HRS/EHRA expert consensus statement on the state of genetic testing for the channelopathies and cardiomyopathies this document was developed as a partnership between the Heart Rhythm Society (HRS) and the European Heart Rhythm Association (EHRA). *Heart Rhythm* **8**, 1308–1339 (2011).
24. Yan, G. X. & Antzelevitch, C. Cellular basis for the Brugada syndrome and other mechanisms of arrhythmogenesis associated with ST-segment elevation. *Circulation* **100**, 1660–1666 (1999).
25. Bloch Thomsen, P. E. *et al.* Phase 2 reentry in man. *Heart Rhythm* **2**, 797–803 (2005).
26. Antzelevitch, C. *In vivo* human demonstration of phase 2 reentry. *Heart Rhythm* **2**, 804–806 (2005).
27. Szel, T. & Antzelevitch, C. Abnormal repolarization as the basis for late potentials and fractionated electrograms recorded from epicardium in experimental models of Brugada syndrome. *J. Am. Coll. Cardiol.* **63**, 2037–2045 (2014).
28. Koncz, I. *et al.* Mechanisms underlying the development of the electrocardiographic and arrhythmic manifestations of early repolarization syndrome. *J. Mol. Cell. Cardiol.* **68**, 20–28 (2014).
29. Patocskai, B. & Antzelevitch, C. Novel therapeutic strategies for the management of ventricular arrhythmias associated with the Brugada syndrome. *Expert Opin. Orphan Drugs* **3**, 633–651 (2015).
30. Coronel, R. *et al.* Right ventricular fibrosis and conduction delay in a patient with clinical signs of Brugada syndrome: a combined electrophysiological, genetic, histopathologic, and computational study. *Circulation* **112**, 2769–2777 (2005).
31. Nademanee, K. *et al.* Prevention of ventricular fibrillation episodes in Brugada syndrome by catheter ablation over the anterior right ventricular outflow tract epicardium. *Circulation* **123**, 1270–1279 (2011).
32. Boukens, B. J. *et al.* Reduced sodium channel function unmasks residual embryonic slow conduction in the adult right ventricular outflow tract. *Circ. Res.* **113**, 137–141 (2013).
33. van Rijen, H. V. *et al.* Slow conduction and enhanced anisotropy increase the propensity for ventricular tachyarrhythmias in adult mice with induced deletion of connexin43. *Circulation* **109**, 1048–1055 (2004).
34. Moorman, A. F. & Christoffels, V. M. Cardiac chamber formation: development, genes, and evolution. *Physiol. Rev.* **83**, 1223–1267 (2003).
35. Borggrefe, M. & Schimpf, R. J-wave syndromes caused by repolarization or depolarization mechanisms a debated issue among experimental and clinical electrophysiologists. *J. Am. Coll. Cardiol.* **55**, 798–800 (2010).
36. Elizari, M. V. *et al.* Abnormal expression of cardiac neural crest cells in heart development: a different hypothesis for the etiopathogenesis of Brugada syndrome. *Heart Rhythm* **4**, 359–365 (2007).
37. Waldo, K. L., Lo, C. W. & Kirby, M. L. Connexin 43 expression reflects neural crest patterns during cardiovascular development. *Dev. Biol.* **208**, 307–323 (1999).
38. Sohl, G. & Willecke, K. Gap junctions and the connexin protein family. *Cardiovasc. Res.* **62**, 228–232 (2004).
39. Kumar, N. M. & Gilula, N. B. The gap junction communication channel. *Cell* **84**, 381–388 (1996).
40. Ewart, J. L. *et al.* Heart and neural tube defects in transgenic mice overexpressing the Cx43 gap junction gene. *Development* **124**, 1281–1292 (1997).
41. Dhar Malhotra, J. *et al.* Characterization of sodium channel alpha- and beta-subunits in rat and mouse cardiac myocytes. *Circulation* **103**, 1303–1310 (2001).
42. Malhotra, J. D., Kazen-Gillespie, K., Hortsch, M. & Isom, L. L. Sodium channel beta subunits mediate homophilic cell adhesion and recruit ankyrin to points of cell-cell contact. *J. Biol. Chem.* **275**, 11383–11388 (2000).
43. Jansen, J. A. *et al.* Reduced heterogeneous expression of Cx43 results in decreased Nav1.5 expression and reduced sodium current that accounts for arrhythmia vulnerability in conditional Cx43 knockout mice. *Heart Rhythm* **9**, 600–607 (2012).
44. Desplantez, T. *et al.* Connexin43 ablation in foetal atrial myocytes decreases electrical coupling, partner connexins, and sodium current. *Cardiovasc. Res.* **94**, 58–65 (2012).
45. Lubkemeier, I. *et al.* Deletion of the last five C-terminal amino acid residues of connexin43 leads to lethal ventricular arrhythmias in mice without affecting coupling via gap junction channels. *Bas. Res. Cardiol.* **108**, 348 (2013).
46. Poelzing, S., Akar, F. G., Baron, E. & Rosenbaum, D. S. Heterogeneous connexin43 expression produces electrophysiological heterogeneities across ventricular wall. *Am. J. Physiol. Heart Circulatory Physiol.* **286**, H2001–2009 (2004).
47. Vaidya, D. *et al.* Null mutation of connexin43 causes slow propagation of ventricular activation in the late stages of mouse embryonic development. *Circ. Res.* **88**, 1196–1202 (2001).
48. Nademanee, K. *et al.* Fibrosis, connexin-43, and conduction abnormalities in the Brugada syndrome. *J. Am. Coll. Cardiol.* **66**, 1976–1986 (2015).
49. Agullo-Pascual, E., Cerrone, M. & Delmar, M. Arrhythmogenic cardiomyopathy and Brugada syndrome: diseases of the connexome. *FEBS Lett.* **588**, 1322–1330 (2014).
50. Rhett, J. M. & Gourdie, R. G. The perinexus: a new feature of Cx43 gap junction organization. *Heart Rhythm* **9**, 619–623 (2012).
51. Petitprez, S. *et al.* SAP97 and dystrophin macromolecular complexes determine two pools of cardiac sodium channels Nav1.5 in cardiomyocytes. *Circ. Res.* **108**, 294–304 (2011).
52. Rhett, J. M., Veeraraghavan, R., Poelzing, S. & Gourdie, R. G. The perinexus: sign-post on the path to a new model of cardiac conduction? *Trends Cardiovascular Med.* **23**, 222–228 (2013).
53. Zareba, W. *et al.* Influence of genotype on the clinical course of the long-QT syndrome. International Long-QT Syndrome Registry Research Group. *N. Engl. J. Med.* **339**, 960–965 (1998).
54. Webster, G. & Berul, C. I. Congenital long-QT syndromes: a clinical and genetic update from infancy through adulthood. *Trends Cardiovascular Med.* **18**, 216–224 (2008).
55. Garcia, J. *et al.* Clinical genetic testing for the cardiomyopathies and arrhythmias: a systematic framework for establishing clinical validity and addressing genotypic and phenotypic heterogeneity. *Front. Cardiovasc.Med.* **3**, 20 (2016).
56. Dendramis, G. Coronary anomalies and Brugada phenocopy, the first documented case in the world. *Int. J. Cardiol.* **199**, 335–336 (2015).
57. Priori, S. G. Natural history of Brugada syndrome: insights for risk stratification and management. *Circulation* **105**, 1342–1347 (2002).
58. Sieira, J. *et al.* Prognostic value of programmed electrical stimulation in Brugada syndrome: 20 years experience. *Circ. Arrhythm. Electrophysiol.* **8**, 777–784 (2015).
59. Probst, V. *et al.* Long-term prognosis of patients diagnosed with Brugada syndrome: results from the FINGER Brugada Syndrome Registry. *Circulation* **121**, 635–643 (2010).
60. Priori, S. G. *et al.* 2015 ESC Guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: the task force for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death of the European Society of Cardiology (ESC). Endorsed by: Association for European Paediatric and Congenital Cardiology (AEPC). *Eur. Heart J.* **36**, 2793–2867 (2015).
61. Priori, S. G. *et al.* HRS/EHRA/APHRS expert consensus statement on the diagnosis and management of patients with inherited primary arrhythmia syndromes: document endorsed by HRS, EHRA, and APHRS in May 2013 and by ACCF, AHA, PACES, and AEPC in June 2013. *Heart Rhythm* **10**, 1932–1963 (2013).
62. Olde Nordkamp, L. R. *et al.* Implantable cardioverter-defibrillator harm in young patients with inherited arrhythmia syndromes: A systematic review and meta-analysis of inappropriate shocks and complications. *Heart Rhythm* **13**, 443–454 (2016).
63. Conte, G. *et al.* Implantable cardioverter-defibrillator therapy in Brugada syndrome: a 20-year single-center experience. *J. Am. Coll. Cardiol.* **65**, 879–888 (2015).
64. Sacher, F. *et al.* Outcome after implantation of a cardioverter-defibrillator in patients with Brugada syndrome: a multicenter study-part 2. *Circulation* **128**, 1739–1747 (2013).
65. Brugada, J. *et al.* Brugada syndrome phenotype elimination by epicardial substrate ablation. *Circ. Arrhythm. Electrophysiol.* **8**, 1373–1381 (2015).
66. Belhassen, B., Rahkovich, M., Michowitz, Y., Glick, A. & Viskin, S. Management of Brugada syndrome: thirty-three-year experience using electrophysiologically guided therapy with class 1A antiarrhythmic drugs. *Circ. Arrhythm. Electrophysiol.* **8**, 1393–1402 (2015).
67. Sroubek, J. *et al.* Programmed ventricular stimulation for risk stratification in the Brugada syndrome: a pooled analysis. *Circulation* **133**, 622–630 (2016).
68. Sieira, J. *et al.* Clinical characterisation and long-term prognosis of women with Brugada syndrome. *Heart* **102**, 452–458 (2016).

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.

69. Bagnall, R. D. *et al.* Prospective study of sudden cardiac death among children and young adults. *N. Engl. J. Med.* **374**, 2441–2452 (2016).
70. Priori, S. G. *et al.* Risk stratification in Brugada syndrome: results of the PRELUDE (PRogrammed ELectrical stimUlation preDictive valuE) registry. *J. Am. Coll. Cardiol.* **59**, 37–45 (2012).
71. Delise, P. *et al.* Risk stratification in individuals with the Brugada type 1 ECG pattern without previous cardiac arrest: usefulness of a combined clinical and electrophysiologic approach. *Eur. Heart J.* **32**, 169–176 (2011).
72. Sieira, J. *et al.* Asymptomatic Brugada syndrome: clinical characterization and long-term prognosis. *Circ. Arrhythm. Electrophysiol.* **8**, 1144–1150 (2015).
73. Brugada, J. Long-term follow-up of individuals with the electrocardiographic pattern of right bundle-branch block and ST-segment elevation in precordial leads V1 to V3. *Circulation* **105**, 73–78 (2002).
74. Deo, R. & Albert, C. M. Epidemiology and genetics of sudden cardiac death. *Circulation* **125**, 620–637 (2012).
75. Delise, P., Allocca, G., Sitta, N. & DiStefano, P. Event rates and risk factors in patients with Brugada syndrome and no prior cardiac arrest: a cumulative analysis of the largest available studies distinguishing ICD-recorded fast ventricular arrhythmias and sudden death. *Heart Rhythm* **11**, 252–258 (2014).
76. Andorin, A. *et al.* Impact of clinical and genetic findings on the management of young patients with Brugada syndrome. *Heart Rhythm* **13**, 1274–1282 (2016).
77. Gonzalez-Corcia, M. *et al.* Brugada syndrome in the young: assessment of risk factors for future events. *Europace* (2016).
78. Conte, G. *et al.* Follow-up from childhood to adulthood of individuals with family history of Brugada syndrome and normal electrocardiograms. *JAMA* **312**, 2039–2041 (2014).
79. Conte, G. *et al.* Clinical characteristics, management, and prognosis of elderly patients with Brugada syndrome. *J. Cardiovasc. Electrophysiol.* **25**, 514–519 (2014).
80. Benito, B. *et al.* Gender differences in clinical manifestations of Brugada syndrome. *J. Am. Coll. Cardiol.* **52**, 1567–1573 (2008).
81. Sacher, F. *et al.* Are women with severely symptomatic Brugada syndrome different from men? *J. Cardiovasc. Electrophysiol.* **19**, 1181–1185 (2008).
82. Sarkozy, A. *et al.* The value of a family history of sudden death in patients with diagnostic type I Brugada ECG pattern. *Eur. Heart J.* **32**, 2153–2160 (2011).
83. Hasdemir, C. *et al.* High prevalence of concealed Brugada syndrome in patients with atrioventricular nodal reentrant tachycardia. *Heart Rhythm* **12**, 1584–1594 (2015).
84. Sommariva, E. *et al.* Genetics can contribute to the prognosis of Brugada syndrome: a pilot model for risk stratification. *Eur. J. Hum. Genet.* **21**, 911–917 (2013).
85. Meregalli, P. G. *et al.* Type of SCN5A mutation determines clinical severity and degree of conduction slowing in loss-of-function sodium channelopathies. *Heart Rhythm* **6**, 341–348 (2009).
86. Richter, S. *et al.* Number of electrocardiogram leads displaying the diagnostic coved-type pattern in Brugada syndrome: a diagnostic consensus criterion to be revised. *Eur. Heart J.* **31**, 1357–1364 (2010).
87. Miyamoto, K. *et al.* Diagnostic and prognostic value of a type 1 Brugada electrocardiogram at higher (third or second) V1 to V2 recording in men with Brugada syndrome. *Am. J. Cardiol.* **99**, 53–57 (2007).
88. Gottschalk, B. H. *et al.* Expert cardiologists cannot distinguish between Brugada phenocopy and Brugada syndrome electrocardiogram patterns. *Europace* **18**, 1095–1100 (2016).
89. Calo, L. *et al. Am. Coll. Cardiol.* **67**, 1427–1440 (2016).
90. Kawata, H. *et al.* Prognostic significance of early repolarization in inferolateral leads in Brugada patients with documented ventricular fibrillation: a novel risk factor for Brugada syndrome with ventricular fibrillation. *Heart Rhythm* **10**, 1161–1168 (2013).
91. Takagi, M. *et al.* The prognostic value of early repolarization (J wave) and ST-segment morphology after J wave in Brugada syndrome: multicenter study in Japan. *Heart Rhythm* **10**, 533–539 (2013).
92. Alings M, Wilde A. "Brugada" syndrome: clinical data and suggested pathophysiological mechanism. *Circulation* **99**, 666–673 (1999).
93. Miyazaki, T. *et al.* Autonomic and antiarrhythmic drug modulation of ST segment elevation in patients with Brugada syndrome. *J. Am. Coll. Cardiol.* **27**, 1061–1070 (1996).
94. Mizumaki, K. *et al.* Vagal activity modulates spontaneous augmentation of ST elevation in the daily life of patients with Brugada syndrome. *J. Cardiovasc. Electrophysiol.* **15**, 667–673 (2004).
95. Viskin, S., Rosso, R., Friedensohn, L., Havakuk, O. & Wilde, A. A. Everybody has Brugada syndrome until proven otherwise? *Heart Rhythm* **12**, 1595–1598 (2015).
96. Havakuk, O. & Viskin, S. A. A tale of 2 diseases: the history of long-QT syndrome and Brugada syndrome. *J. Am. College Cardiol.* **67**, 100–108 (2016).
97. Brugada, R. *et al.* Sodium channel blockers identify risk for sudden death in patients with ST-segment elevation and right bundle branch block but structurally normal hearts. *Circulation* **101**, 510–515 (2000).
98. Morita, H. *et al.* Fragmented QRS as a marker of conduction abnormality and a predictor of prognosis of Brugada syndrome. *Circulation* **118**, 1697–1704 (2008).
99. Okamura, H. *et al.* Risk stratification in patients with Brugada syndrome without previous cardiac arrest - prognostic value of combined risk factors. *Circ. J.* **79**, 310–317 (2015).
100. Sarkozy, A. *et al.* Inferior and lateral electrocardiographic repolarization abnormalities in Brugada syndrome. *Circ. Arrhythm. Electrophysiol.* **2**, 154–161 (2009).
101. Kamakura, T. *et al.* Significance of electrocardiogram recording in high intercostal spaces in patients with early repolarization syndrome. *Eur. Heart J.* **37**, 630–637 (2016).
102. Zumhagen, S. *et al.* Tpeak-Tend interval and Tpeak-Tend/QT ratio in patients with Brugada syndrome. *Europace* http://dx.doi.org/10.1093/europace/euw033 (2016).
103. Castro Hevia, J. *et al.* Tpeak-Tend and Tpeak-Tend dispersion as risk factors for ventricular tachycardia/ventricular fibrillation in patients with the Brugada syndrome. *J. Am. Coll. Cardiol.* **47**, 1828–1834 (2006).
104. Babai Bigi, M. A., Aslani, A. & Shahrzad, S. aVR sign as a risk factor for life-threatening arrhythmic events in patients with Brugada syndrome. *Heart Rhythm* **4**, 1009–1012 (2007).
105. Uchimura-Makita, Y. *et al.* Time-domain T-wave alternans is strongly associated with a history of ventricular fibrillation in patients with Brugada syndrome. *J. Cardiovasc. Electrophysiol.* **25**, 1021–1027 (2014).
106. Morita, H. *et al.* Atrial fibrillation and atrial vulnerability in patients with Brugada syndrome. *J. Am. Coll. Cardiol.* **40**, 1437–1444 (2002).
107. Rodriguez-Manero, M. *et al.* Prevalence, clinical characteristics and management of atrial fibrillation in patients with Brugada syndrome. *Am. J. Cardiol.* **111**, 362–367 (2013).
108. Giustetto, C. *et al.* Atrial fibrillation in a large population with Brugada electrocardiographic pattern: prevalence, management, and correlation with prognosis. *Heart Rhythm* **11**, 259–265 (2014).
109. Morita, H. *et al.* Sinus node function in patients with Brugada-type ECG. *Circ. J.* **68**, 473–476 (2004).
110. Letsas, K. P. *et al.* Sinus node disease in subjects with type 1 ECG pattern of Brugada syndrome. *J. Cardiol.* **61**, 227–231 (2013).
111. Abe, K. *et al.* Sodium channelopathy underlying familial sick sinus syndrome with early onset and predominantly male characteristics. *Circ. Arrhythm. Electrophysiol.* **7**, 511–517 (201 4).
112. Brugada, P., Geelen, P., Brugada, R., Mont, L. & Brugada, J. Prognostic value of electrophysiologic investigations in Brugada syndrome. *J. Cardiovasc. Electrophysiol.* **12**, 1004–1007 (2001).
113. Eckardt, L. *et al.* Electrophysiologic investigation in Brugada syndrome; yield of programmed ventricular stimulation at two ventricular sites with up to three premature beats. *Eur. Heart J.* **23**, 1394–1401 (2002).
114. Takigawa, M. *et al.* Seasonal and circadian distributions of ventricular fibrillation in patients with Brugada syndrome. *Heart Rhythm* **5**, 1523–1527 (2008).
115. Stroker, E., de Asmundis, C., Chierchia, G. B. & Brugada, P. Exercise-related Brugada pattern and monomorphic ventricular tachycardia in a patient with Brugada syndrome: interplay between body temperature, haemodynamics and vagal activity. *Eur. Heart J.* **37**, 655 (2016).
116. Belhassen, B., Glick, A. & Viskin, S. Efficacy of quinidine in high-risk patients with Brugada syndrome. *Circulation* **110**, 1731–1737 (2004).
117. Anguera, I. *et al.* Shock reduction with long-term quinidine in patients with Brugada syndrome and malignant ventricular arrhythmia episodes. *J. Am. Coll. Cardiol.* **67**, 1653–1654 (2016).
118. Eckardt, L. *et al.* Long-term prognosis of individuals with right precordial ST-segment-elevation Brugada syndrome. *Circulation* **111**, 257–263 (2005).
119. Takagi, M., Yokoyama, Y., Aonuma, K., Aihara, N. & Hiraoka, M. Clinical characteristics and risk stratification in symptomatic and asymptomatic patients with brugada syndrome: multicenter study in Japan. *J. Cardiovasc. Electrophysiol.* **18**, 1244–1251 (2007).
120. Brugada, J., Brugada, R. & Brugada, P. Determinants of sudden cardiac death in individuals with the electrocardiographic pattern of Brugada syndrome and no previous cardiac arrest. *Circulation* **108**, 3092–3096 (2003).

Author contributions

J.S. and G.D. contributed equally to this work. All the authors researched data for the article, substantially contributed to discussion of content, and wrote, reviewed, and edited the manuscript before submission.

Competing interests statement

P.B. has received institutional grants from Biotronik, Medtronic, and St. Jude Medical. The other authors declare no competing interests.

© 2016 Macmillan Publishers Limited, part of Springer Nature. All rights reserved.